Alok Kadam

Embedded Firmware Engineer

RESUME_1766

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Total Experience: 4 years

PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 4 years of experience in embedded firmware, using C, C++, ARM Cortex-M, FreeRTOS. Experienced in real-time systems, hardware debugging and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: UART, JTAG, ARM Cortex-M, I2C, FreeRTOS, Firmware Debugging, C++, SPI, STM32, Git, Oscilloscope, C

Methods: Real-time systems, Hardware debugging, Protocol validation

PROFESSIONAL EXPERIENCE

Embedded Firmware Engineer | HelioNex Labs | Jan 2024 - Present

- Validated 831 hardware test vectors across protocol and timing corner cases
- Reduced embedded task jitter by 14% using interrupt priority and queue tuning
- Lowered field-reset incidents by 38% with watchdog recovery and diagnostics
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Associate Embedded Firmware Engineer | Meridian Vector Labs | Jul 2022 - Dec 2023

- Lowered field-reset incidents by 12% with watchdog recovery and diagnostics
- Improved battery runtime by 15% with peripheral and sleep-state optimization

PROJECTS

Low-Power Metering Device | STM32, ARM Cortex-M, I2C, JTAG

- Optimized interrupt-driven acquisition and sleep cycles.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

Industrial Sensor Firmware | SPI, FreeRTOS, STM32, C++

- Built FreeRTOS tasks for calibrated sampling and watchdog recovery.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

EDUCATION

B.Tech in Electronics and Communication Engineering, Western Plains Institute of Technology, 2022
Relevant Methods: Real-time systems, Hardware debugging, Protocol validation

CERTIFICATIONS

RTOS Scheduling Training (2026)